

YUMENZHEN, China

WMO: 524360

Lat: 40.2695N

Lon: 97.0271E

Elev: 1527

StdP: 84.28

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.3	-17.9	-27.9	0.3	-12.0	-25.9	0.4	-10.4	10.4	-5.9	9.3	-6.0	2.4	90	0.525

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.9	32.2	16.0	30.9	15.4	29.6	14.9	17.9	27.1	17.1	26.6	16.3	25.7	3.3	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.4	13.1	20.1	14.3	12.2	19.8	13.1	11.3	19.9	57.2	27.3	54.1	26.3	51.7	25.9	23.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
9.5	8.3	7.2		-23.7	34.8	2.8	1.2	-25.7	35.7	-27.3	36.4	-28.9	37.1	-30.9	38.0	
				-24.1	19.8	2.6	1.3	-26.0	20.8	-27.5	21.5	-29.0	22.3	-30.9	23.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.3	-9.1	-4.3	3.0	10.7	16.1	20.6	22.6	21.1	15.8	8.5	0.6	-7.2
	DBStd	11.58	4.20	4.43	4.90	4.60	3.46	2.55	2.41	2.51	3.22	3.75	4.39	4.25
	HDD10.0	2150	593	399	219	46	3	0	0	0	2	72	282	533
	HDD18.3	4004	851	633	475	232	83	9	1	6	87	304	532	792
	CDD10.0	1516	0	0	3	66	193	318	389	345	176	27	0	0
	CDD18.3	330	0	0	0	2	15	77	132	93	11	0	0	0
	CDH23.3	3935	0	0	0	37	268	887	1486	1073	182	2	0	0
	CDH26.7	1268	0	0	0	6	43	251	580	361	28	0	0	0
Wind	WSAvg	3.2	3.5	3.5	3.8	3.9	3.1	2.7	2.6	2.6	2.6	2.8	3.5	3.6
Precipitation	PrecAvg	59	1	2	4	5	8	11	14	10	5	2	2	1
	PrecMax	93	5	8	15	18	40	39	68	35	21	14	11	4
	PrecMin	25	0	0	0	0	0	0	2	0	0	0	0	0
	PrecStd	20	1	2	4	5	11	11	14	9	6	3	3	1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.3	10.6	19.4	27.0	29.7	32.3	34.4	33.6	29.4	23.4	15.0	6.3
		MCWB	-1.5	1.2	5.9	10.0	12.5	15.1	17.2	16.3	13.5	9.5	4.4	-0.9
	2%	DB	2.0	7.6	16.6	23.8	28.0	30.7	32.7	31.7	27.2	20.8	12.5	3.4
		MCWB	-3.5	-0.3	4.7	8.7	11.7	14.5	16.5	15.6	12.4	8.6	3.3	-2.4
	5%	DB	-0.3	5.6	14.2	21.6	26.4	29.5	31.4	30.2	25.5	18.6	10.5	1.2
		MCWB	-4.8	-1.2	3.5	7.6	10.9	14.1	16.2	15.1	11.8	7.5	2.3	-3.6
	10%	DB	-2.3	3.6	12.0	19.7	24.6	28.1	30.0	28.8	23.8	16.5	8.2	-0.7
		MCWB	-6.1	-2.4	2.5	6.8	10.4	13.8	15.7	14.8	11.4	6.3	1.2	-4.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.3	1.6	6.3	10.8	13.3	17.6	20.1	19.0	15.6	10.6	5.2	-0.5
		MCDB	4.9	9.7	18.3	25.1	27.2	26.4	30.2	27.5	22.3	19.3	13.5	5.7
	2%	WB	-3.3	0.0	4.9	9.1	12.4	16.6	18.4	17.5	14.0	9.1	3.6	-2.2
		MCDB	1.9	7.1	15.7	22.4	25.5	25.8	28.1	26.4	23.1	19.6	11.9	2.9
	5%	WB	-4.7	-1.1	3.8	8.1	11.6	15.7	17.5	16.6	13.0	8.0	2.5	-3.4
		MCDB	-0.5	5.3	13.6	20.4	24.2	25.2	27.2	25.8	22.6	17.8	10.0	0.9
	10%	WB	-6.0	-2.2	2.7	7.1	10.8	14.9	16.7	15.9	12.1	6.8	1.3	-4.6
		MCDB	-2.5	3.3	11.5	18.5	23.0	24.8	26.5	25.2	21.5	15.7	8.0	-0.9
Mean Daily Temperature Range	5% DB	MDBR	11.2	12.9	13.6	13.5	13.4	12.8	12.9	13.5	13.9	13.6	11.2	10.0
		MCDBR	13.4	15.3	16.7	16.3	16.4	15.6	15.9	16.6	16.3	16.3	13.8	12.2
		MCWBR	9.4	9.5	8.8	7.3	6.9	6.3	5.8	5.7	6.2	7.8	7.8	8.1
	5% WB	MCDBR	13.2	14.5	15.7	14.8	14.2	12.6	12.7	13.1	13.2	15.3	13.3	11.7
		MCWBR	9.3	9.0	8.3	6.7	6.2	5.5	5.3	5.0	5.3	7.4	7.6	7.9
Clear-Sky Solar Irradiance	taub	0.300	0.350	0.411	0.465	0.466	0.480	0.468	0.449	0.420	0.372	0.328	0.300	
	taud	2.208	2.071	1.920	1.821	1.876	1.910	2.001	2.031	2.047	2.090	2.158	2.209	
	Ebn at Noon	847	844	827	809	818	806	811	815	810	807	797	811	
	Edh at Noon	98	132	172	204	198	191	173	163	150	128	103	91	
All-Sky Solar Radiation	RadAvg	2.64	3.63	5.00	6.18	6.96	6.97	6.68	6.18	5.35	4.28	2.96	2.33	
	RadStd	0.15	0.17	0.27	0.35	0.34	0.52	0.27	0.36	0.32	0.16	0.13	0.15	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	+79	+36

Nomenclature: See separate page